UDCT v7.12 Preregistration

Anisotropy and Screening-Length Boundaries of Signed Jeans Solutions in Hydrus I

Won Shik Paik
Independent Researcher | Auckland, New Zealand

Status and primary question

Prepared after the v7.11 results were known. This is a prospective follow-up of new parameter values, not an independent blind prediction of the earlier negative cases. No v7.12 galaxy cases were evaluated during preparation. Only synthetic solver, decision, bracket and serialization tests were run.

Public preregistration is incomplete until the protocol and exact source package have been publicly deposited with a timestamp. No v7.12 DOI or repository URL is claimed in this document. Those identifiers will be recorded later; a reserved DOI alone is not publication evidence.

The primary question is where the prescribed stationary spherical-average Jeans closure changes its sampled sign as constant velocity anisotropy β and screening length l_0 vary. The study maps numerical N/P/I regions and resolves only adequately supported zero brackets.

Known evidence and expectations

v7.11 reports N for every tested screened cusp case at 3 and 10 pc. At 1 pc and $\beta=0.5$, BASE, A-f2, A-Qs and A-Qd are P; A-f1/2 is I. These are known motivating results. They do not guarantee a unique transition, monotonicity, or a positive endpoint for A-f1/2.

Directional expectation: some 1 pc laws may possess a sign boundary within $0\le\beta\le0.5$, and that boundary may shift over $1\le l_0\le3$ pc. No particular critical value is predicted. Absent, multiple and numerically unresolved boundaries are valid outcomes.

Interpretation

P is a positive sampled second-moment candidate, not a physical equilibrium, positive distribution function, stability proof or observational success. N is an obstruction under this closure and boundary condition, not simulated dynamical collapse. Neither outcome validates UDCT or excludes MOND generally.

1. Frozen design and execution order

Reuse the unchanged v7.11 signed Jeans adapter and the hash-locked v7.10/v7.06 gravity sources. Hydrus I stellar inputs, mass normalization, baseline r_h, Galactic acceleration and Hessian, force convention, action-consistent gravity and numerical grids are inherited without fitting.

Component	Registered specification
Density	Unsoftened Tapered cusp only; ε=0. Plummer is not part of this new matrix.
Screened laws	BASE, A-f1/2, A-f2, A-Qs, A-Qd
β values	0, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50
l_0 values (pc)	1, 1.2, 1.5, 2, 2.5, 3
Controls	QUMOND-nu0, nu1, nu2, each at all 11 β values; screening length absent
Grids / inner limits	smoke, default, refined; r_{in}/r_h=0.002 and 0.0005
Outer boundary	R_{out}=40 r_h; P_{out}=0
Primary radii	1,200 common logarithmic points from 0.002 to 10 r_h; valid native minima retained

Stage 1: anisotropy at 1 pc

Evaluate all five screened laws at l_0=1 pc, plus all three QUMOND controls, at the eleven β values. Complete all three grids and both inner limits: 48 field evaluations, 528 pressure records, 88 primary decisions.

Stage 2: screening-length map

Evaluate the remaining five l_0 values for every screened law at all eleven β values: 150 field evaluations, 1,650 pressure records, 275 primary decisions. Controls are not duplicated by length. Both fixed stages finish before adaptive refinement.

Fixed total: 198 field evaluations, 2,178 pressure records and 363 primary decisions. Previously known endpoint settings are rerun as anchors; they are not described as new blind evidence.

Eligibility and deferred extensions

Cusp β≤0.5 passes the inherited necessary central-slope screen only; applicability depends on the potential and does not establish distribution-function positivity. Core softening and local-u analysis are deferred, requiring their own specification before affected calculations. No density adjustment is made to improve an outcome.

2. Signed solver and classification

Use $P_r=\rho_\star\sigma_r^2$, positive inward acceleration g_{in}, and constant $\beta=1-(\sigma\theta^2+\sigma\varphi^2)/(2\sigma_r^2)$. The unchanged solver evaluates the signed cumulative trapezoidal integral inward:

$$dP_r/dr + 2\beta P_r/r = -\rho_\star g_{in}$$
$$q(r) = (r/R_{out})^{2\beta}$$
$$I(r) = \int_r^{R_{out}} q(s)\rho_\star(s)g_{in}(s)\,ds$$
$$P_r(r) = [P_{out}+I(r)]/q(r), \text{ with } P_{out}=0$$

Negative pressure and variance remain signed. No clipping, absolute-value conversion or square-root replacement is permitted. Cells with $\rho_\star \le 10^{-280}$ are excluded by the frozen underflow convention. Normalization is $v_{ref}^2=r_h g_{N,\star}(r_h)$.

Gate R: mandatory isotropic reproduction

At each distinct field evaluation, $\beta=0$ must reproduce the inherited isotropic variance to maximum absolute error/$v_{ref}^2 \le 10^{-10}$. Missing or modified frozen sources abort execution. A failed gate stops the attempt, preserves its journal and suppresses outcome claims.

N/P/I rules inherited from v7.11

On common radii, let a,b,c,d be normalized variance arrays for default/0.002, refined/0.002, default/0.0005 and refined/0.0005. Define $E=\max(\sup|a-b|, \sup|c-d|, \sup|b-d|)$ and $\tau=\max(10^{-8},5E)$. This is an empirical numerical diagnostic, not a rigorous error bound.

Label	Rule
N	At least one common radius has all four values below $-\tau$.
P	All four common arrays and each corresponding valid native minimum up to 10 r_h exceed τ.
I	All other cases, including exact zero, inconsistent comparisons and insufficient numerical margin.

Smoke runs are audits, not deciding comparisons. A numerical $F<-\tau$ (defined on the next page) is not automatically N: a negative native minimum may lack a common overlapping negative band. The full N/P/I decision is always retained separately.

Preserved diagnostics

Retain minimum values and radii, inner-endpoint minima, sampled negative intervals, original global work, required outer-pressure thresholds and the finite-shell virial residual. These diagnostics cannot establish external confinement or full three-dimensional equilibrium. Fixed density and gravity imply global work is independent of β.

3. Pre-specified bracket refinement

Continuous observable and numerical margin

For each evaluated (law,l_0,β), define F as the smallest normalized σ_r^2 value over the four deciding common arrays and their valid native minima up to 10 r_h. This lower envelope is a sampled scalar used for zero-bracket refinement, not a physical stability parameter.

Raw sign is the sign of F. Numerical sign is -1 for F<-τ, +1 for F>τ and 0 otherwise. Refinement begins only for adjacent fixed-grid endpoints whose finite F values have opposite raw signs. Endpoints inside the margin leave that bracket unresolved without adding a midpoint.

Deterministic search

After completing both fixed stages, enumerate every adjacent β pair for all 30 screened law/length settings (300 possible pairs), then every adjacent l_0 pair for all 55 screened law/β settings (275 possible pairs). QUMOND controls remain fixed-grid controls and are not adaptively refined.

For each original strict sign-changing pair, bisect the coordinate interval. Evaluate all six grid/inner combinations at every new point, apply Gate R to every new field, and retain the complete decision. Reuse exact cached fields and previously evaluated coordinates.

Stop rule	Action
β width ≤0.001	Report the remaining bracket and endpoint errors, not an exact βcrit.
l_0 width ≤0.01 pc	Report the remaining bracket and endpoint errors, not an exact l_0,crit.
Midpoint |F|≤τ or invalid comparison	Stop this bracket as unresolved; preserve endpoints and midpoint.
20 midpoint steps	Stop with iteration-limit status; retain the remaining interval.

At most 575 original pairs and 11,500 midpoint evaluations are permitted by this conservative cap. No outcome-driven expansion beyond the registered ranges. Multiple roots within one bracket remain possible; no monotonicity or uniqueness is presumed. Tangential zeros without sampled sign changes may be missed and must be acknowledged.

Required reporting

Publish the complete fixed N/P/I map, continuous F and numerical margins, all sampled-zero cases, all attempted brackets, unresolved intervals and failures. A robust N-to-P transition and a raw F-zero bracket are reported as distinct findings. No detected bracket means only none was detected within the prescribed samples and assumptions.

A root of the integrated Jeans solution is not necessarily a local acceleration-zero radius. β changes Jeans weighting while the frozen gravitational field and Hessian remain unchanged. No universal ucrit or Hessian threshold is inferred from this experiment.

4. Validation, publication and provenance

The preparation self-test checks the frozen-source hashes; inherited analytic constant-force solutions; every new β against an analytic solution; signed negative-pressure and boundary behavior; exact-zero and native-minimum decisions; analytic bisection and uncertainty stopping; and compact gzip JSON round-trip. No galaxy or Galactic-field evaluations are performed by --selftest.

Package and preparation commands

Keep the v7.12 runner and the three frozen Python files in one directory. The source archive contains these files, requirements.txt, SHA256SUMS.txt and this PDF. README is intentionally deferred until the public GitHub and Zenodo links exist.

```
python -m pip install -r requirements.txt
python UDCT_v7_12_Preregistration_Jeans_Transition.py
python UDCT_v7_12_Preregistration_Jeans_Transition.py --selftest
```

Tested preparation environment: Python 3.12 series, NumPy 2.3.5, SciPy 1.17.0. The requirements pin NumPy and SciPy. Record the actual installed versions at execution. Passing synthetic tests does not certify a completed galaxy run.

Before new galaxy execution

Publicly publish the exact PDF and source package, retain the publication timestamp, and record the real registration URL. Then invoke --run with --registration-url, --registration-published-utc, --confirm-public-registration and --output ending in .json.gz. The UTC timestamp must be in the past. The operator attests publication; the runner does not independently retrieve or verify the URL.

The runner records the command, UTC start/completion, publication attestation, environment and source hashes. It writes an incremental compressed journal and a compact compressed result. It reopens and parses the complete result and checks the 363 fixed decisions before installing the final filename without overwriting an existing file.

Failure and deviation rules

Preserve all attempt files, journals and failures. A new attempt requires a new output filename. No optional stopping based on favorable results or removal of unfavorable cases is allowed. Resource interruption permits only an explicitly partial report. This runner does not automatically resume interrupted jobs.

Corrections that can affect registered quantities require a public timestamped amendment before affected cases are rerun. Preserve both versions and explain the change. Changes limited to links or README must not silently alter the locked scientific source package.

Reporting boundary

This document contains no v7.12 galaxy outcomes. A later report must distinguish known v7.11 evidence, the v7.12 fixed matrix, its pre-specified adaptive refinement and any subsequent exploratory work. Non-sphericity, LMC/SMC fields, variable β(r), core changes, full distribution functions and observed-velocity fits are outside scope.

5. Source lock and references

The new runner verifies all three frozen source hashes before any galaxy execution. The delivered PDF records the exact new runner hash below. Archive SHA256SUMS.txt also locks the PDF and requirements file. No archive checksum is embedded in the archive itself.

UDCT_v7_12_Preregistration_Jeans_Transition.py

3b5f93766940243d257f865691f708fd
904b2ba9384003ae407f911179059b49

UDCT_v7_11_Preregistration_Anisotropic_Jeans.py

3f00abae251d7be64a4ca5abb48b4f55
0644b21c946206571fd448464b13ade2

UDCT_v7_10_Postregistration_run.py

a4d1a07ebcaa06348be3022d0f1f6ca4
63347e00194f7fde59f2bd9dbcd521b7

UDCT_v7_06_Hessian_Term_Sign_Audit_reproduce.py

a64440e1350ae8bf10eb3188ec34829b
6d586abf275de5cdc841a50ad3478284

Public antecedents

[1] Paik, W. S. (2026). UDCT v7.11 preregistration: Constant-Anisotropy Jeans and Outer-Boundary Sensitivity in Hydrus I. https://doi.org/10.5281/zenodo.23244522

[2] Paik, W. S. (2026). UDCT v7.11 post-registration results. https://doi.org/10.5281/zenodo.23263099
Results repository: github.com/wspaik5/UDCT_v7.11_Postregistration_Results

[3] Paik, W. S. (2026). UDCT v7.10 post-registration results. https://doi.org/10.5281/zenodo.23233837

[4] Milgrom, M. (2023). Generalizations of quasilinear MOND. Physical Review D 108, 084005. Direct theoretical precedent for the higher-derivative GQUMOND class. https://arxiv.org/abs/2305.01589

[5] An, J. & Evans, N. W. (2006). A Cusp Slope - Central Anisotropy Theorem. Astrophysical Journal 642, 752-758. https://arxiv.org/abs/astro-ph/0511686

[6] Mamon, G. A. & Boué, G. (2010). Kinematic deprojection and mass inversion of spherical systems of known velocity anisotropy. MNRAS 401, 2433-2450. https://arxiv.org/abs/0906.4971

Preparation date: 10 October 2026, Auckland. Public registration date and identifiers are not yet assigned. No new galaxy results are included.